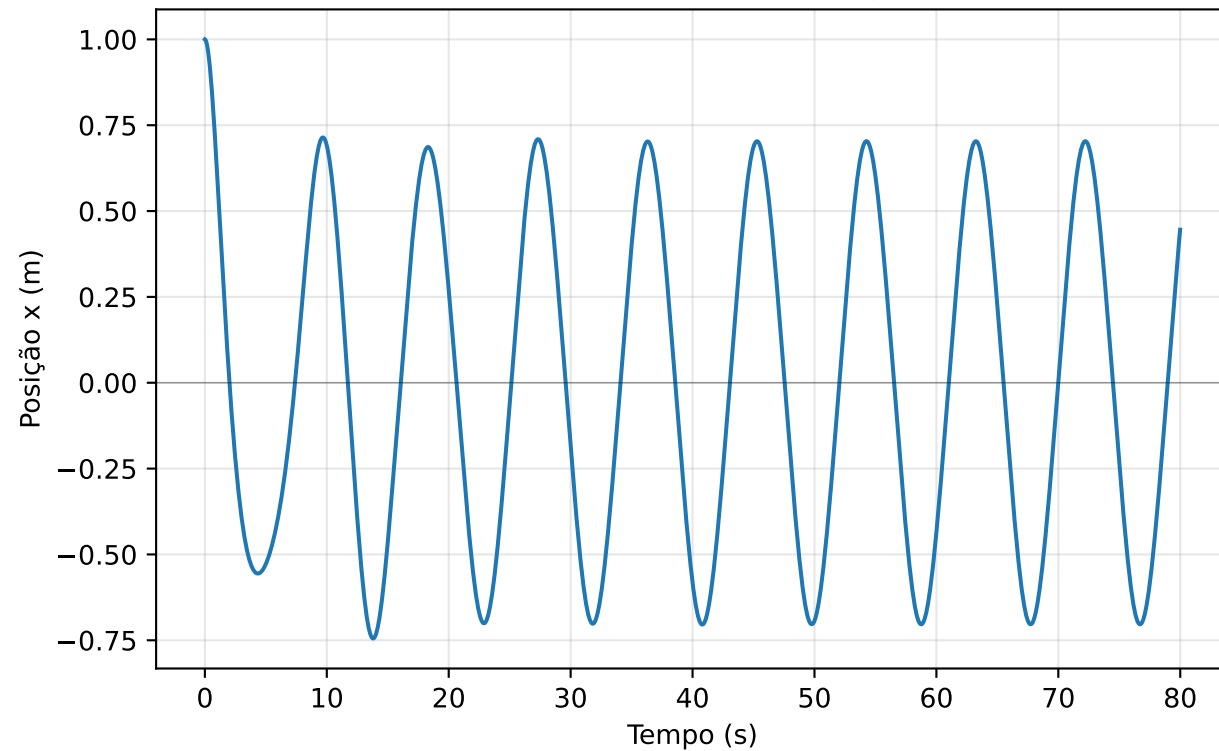
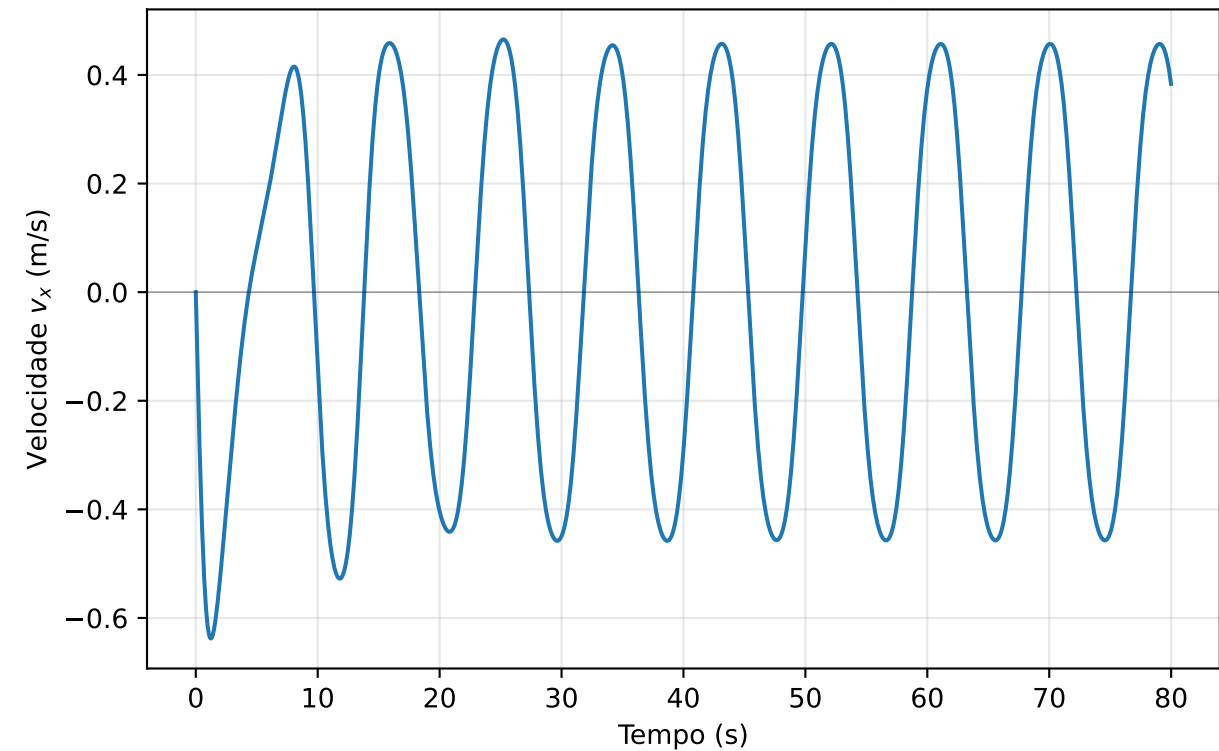


Questão 3 — Simulação base ($k = 1.0$, $\alpha = 0.5$, $b = 0.3$, $F_0 = 0.5$, $\omega = 0.7$ rad/s)

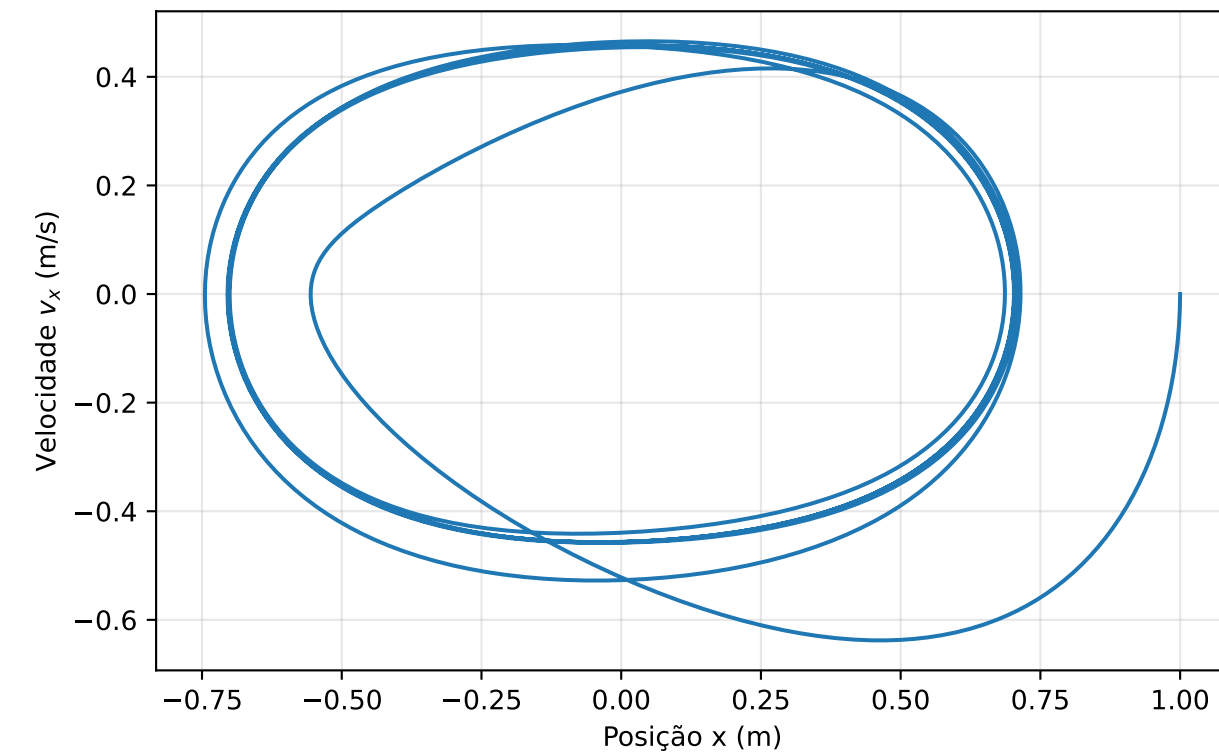
Posição ao longo do tempo



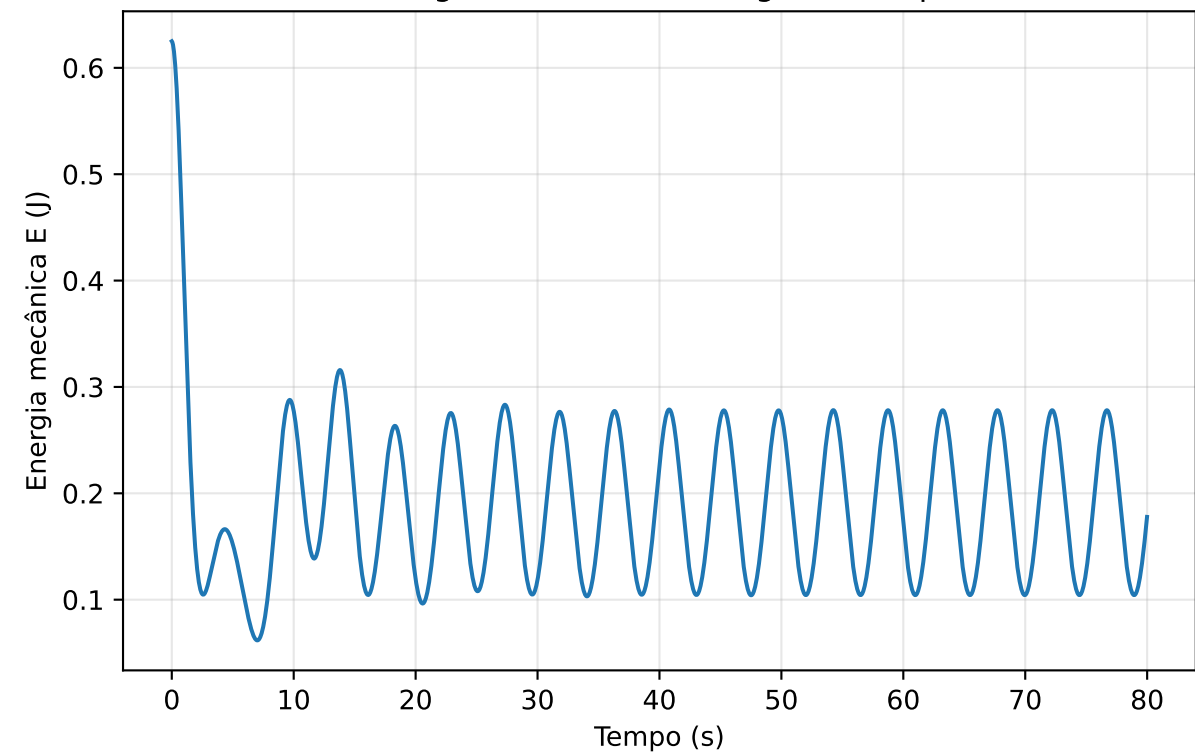
Velocidade ao longo do tempo



Retrato de fase



Energia mecânica ao longo do tempo



$k = 1.0$, $\alpha = 0.5$, $b = 0.3$, $F_0 = 0.5$, $\omega = 0.7$